Five Languages, Five Bottlenecks: Diacritization and G2P for Transliteration at Interscript

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Abstract

We modernized the phonological layer of the Interscript transliteration platform across five languages: Arabic, Hebrew, Thai, Persian, and Urdu, covering both diacritization (vowel restoration in the original script) and G2P (romanization/phonemization). Our central finding is a *bottleneck taxonomy*: each language's performance ceiling was set by exactly one resource—data *quality* (Arabic), data *domain* (Hebrew), labeled-data *coverage* (Thai), output *representation* (Persian, Urdu)—and a fixed ByT5-based framework with one surgical fix per language reached or beat the published state of the art in each. Results: Arabic **2.29% DER** (CE) / 1.33% (w/o CE) on the SadeedDiac-25 benchmark with Misraj's own evaluator under a zero-skip windowed protocol—3.2×/4.0× better than the 1.5B-parameter Sadeed model, ahead of GPT-4 and Gemini-Flash-2.0—plus a frontier regression axis: GLM-5.2 matches our dedicated 580M teacher's vowel accuracy to within 0.02pp, while every GLM successor we measured is 3–5× worse (5.3-Flash 8.80, 5.3 9.90, 4.7-flash 13.23 zero-skip), the loss located by per-position attribution in wrong haraqat, not writing convention; a 300M distilled client tier at 4.82% (42% error reduction) whose lever ladder brackets two registered negatives (MTP-auxiliary, register swap) mirroring the three-way RL negative (RAFT, sequence-GRPO, entropy-weighted GRPO); a contamination audit showing the benchmark's public source corpus leaks 122 of its own paragraphs; Hebrew 16.4% DER on Biblical text where the SOTA model scores 35.6%, and strong modern/poetry/rabbinic line-level surfaces (0.52/0.24/0.45 DER); Thai 2.32% PER via deterministic (non-LLM) augmentation; Persian homograph accuracy 77.3–89.5% and diacritization 0.52% CER from a G2P-only dataset; Urdu G2P 14.77% CER (first learned baseline, 4.1× over rule-based) and diacritization 3.74% CER from weak supervision. We distill cross-cutting lessons for phonological NLP: LLMs are unreliable phonological labelers and are *regressing* on classical knowledge as their optimization shifts agentic; evaluation domains and input formats silently dominate comparisons; ensembling and curricula fail at data-limited optima; policy optimization adds nothing at SFT convergence—every verified gain came from data or supervision curation; and byte-level models are the right deployment target for multilingual TS runtimes.

1 Why phonology for transliteration?

Transliteration maps (ALA-LC, SBL, RTGS, ...) assume *vocalized* input. Undiacritized Arabic or Hebrew, or unsegmented Thai, cannot be transliterated unambiguously. The models described here supply the vocalization that makes interscript.org's maps deterministic.

2 System and results

Common framework: ByT5 seq2seq (byte-level; tokenizer = UTF-8 bytes, so TS inference needs no vocabulary assets) except Arabic, where a compact 30M char-level encoder won on cost.

Language	Task	Result	Published reference
Arabic	diacritization	**2.29/1.33% DER**	Sadeed 7.29/5.26%, Gemini 3.19/2.38%, GLM-5.3-Flash 8.80 (sam
Hebrew	diacritization	**16.43% DER** (Biblical)	DictaBERT 35.6% (same test)
Thai	G2P	**2.32% PER**	baseline 6.37% (same test)
Persian	G2P/HA	**89.5%** / 77.3% (SB)	Homo-GE2PE 76.9%
Persian	diacritization	**0.52% CER**	(none existed)
Urdu	G2P	**14.77% CER**	epitran 60.0%
Urdu	diacritization	**3.74% CER**	(none existed)

3 The bottleneck taxonomy

- **Arabic — data quality**: cleaning + 28× scaling halved DER; a 30M encoder matches a 1.5B LM; supervision-side moves (paragraph context, morphological auxiliary, teacher labeled domain mix) then carried the 580M teacher from 2.81 to 2.29%; a distilled 300M client tier reached 4.82% (42% error reduction) through a controlled lever ladder whose two registered negatives (MTP-auxiliary +0.26pp, register swap +0.98pp) bracket it—supervision quality, not technique, moves this task.
- **Hebrew — data domain**: the SOTA model collapses cross-domain (9×); mixed-domain training wins; the teamim input-format effect makes cantillation copy-through; modern, poetry, and rabbinic line-level surfaces are strong (0.52/0.24/0.45 DER on the public-domain Dicta ACL-2020 test corpora), so the residual is specifically paragraph-level Biblical text.
- **Thai — labeled coverage**: 10K dictionary → 60K via deterministic phonemizer labels; all architectural fixes failed, data fixed it.
- **Persian — representation**: one dataset serves two tasks (G2P + diacritization); metric representation choices (ezafe) shift results by 18 points.
- **Urdu — supervision existence**: no gold diacritization exists; weak IPA-derived labels at 597K scale suffice.

4 Cross-cutting lessons

1. **LLMs cannot label phonology—and are regressing on classical knowledge.** Tone and haraqat hallucination mirrors the known diacritization failure; epitran labels are free of it and transfer through fine-tuning. On SadeedDiac-25 the GLM family regressed 3–5× across one generation boundary (5.2 at 2.69% zero-skip vs 5.3-Flash 8.80, 5.3 9.90, 4.7-flash 13.23), with the loss attributed to wrong haraqat rather than convention—dedicated distilled models are not nostalgia; they are the only reproducible, protocol-pinned artifacts on the right side of that axis.
2. **Input formats and domains are hidden variables.** Teamim-preserving inputs and Biblical-vs-modern test domain each move DER by more than most modeling choices.
3. **Beam search is a first-class factor** (12 DER points on Hebrew); ensembles must be evaluated under beam search to be meaningful.
4. **Output-vote ensembling hurts seq2seq restoration**: per-character voting lowers error counts yet raises edit-distance metrics (incoherent splices).

5. **Curricula fail at data-limited optima**; their reported gains live in pretraining regimes.
6. **Byte-level models are deployment-friendly**: ByT5's tokenizer is `TextEncoder` in TypeScript—no vocab files, no WASM sentencepiece—ideal for interscript.org's browser/Node targets. Quantization fragility is diagnosable and fixable at the artifact level: the quantized output head, not the body, caused confident flip errors (9.34% $\rightarrow$ 0.26% after the head-fp32 repair).

5 Deployment

Best checkpoints export to ONNX; distribution follows the Interscript plan (GitHub Releases primary, HuggingFace canonical, jsDelivr edge). The TS package wraps onnxruntime-web/node with byte-level tokenization.

6 Reproducibility

Per-language repositories (`rababa`, `rababa-farsi`, `rababa-urdu`, `secryst`) contain training pipelines, evaluation harnesses, `docs/RESULTS.md` ground truth, and the five companion papers in `docs/paper-*/`.